

AI-POWERED LANDSLIDE ASSESSMENT REPORT

Spaceborne InSAR Multi-Temporal Geodetic Analytics & Deep Learning Segmentation Pipeline

Geohazard Research & Development Division

Report / Sample ID:	sample_00001
Image ID:	2_56
Region:	Vanoise
Temporal Baseline:	12days
Dataset Split:	Test
Patch Number:	coherence
Report Date:	2026-07-06
Report Time:	12:43:24

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1. Executive Summary

Slope stability diagnostic assessment compiled for site **Vanoise** (Assessment ID: sample_00001). Based on the spaceborne InSAR and deep learning segmentation analysis, the slope is categorized under a **LOW** risk rating with a severity index of **0.2975964407383719** and a system confidence level of **Low**. The active segmented landslide core covers **896 px** (8.96% of the monitored frame).

2. Spatial Mapping Viewport (Visual Pipeline)

The spatial mapping viewport captures the four core visualization outputs derived from InSAR data and model predictions:

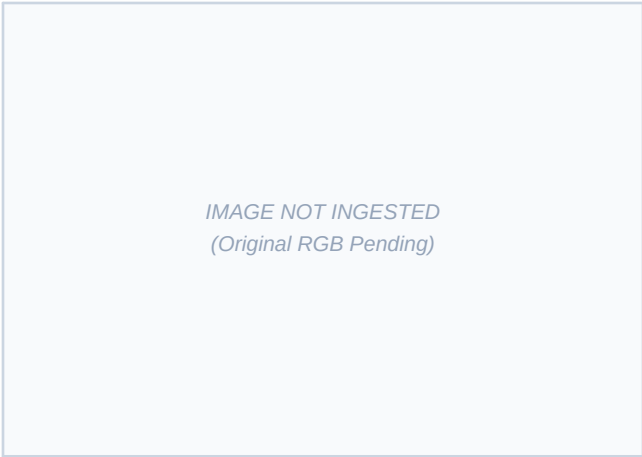


Fig 1: Original InSAR RGB
R=Coherence, G=cos(Phase), B=sin(Phase)



Fig 2: SegFormer Prediction Mask
Binary landslide classification boundary



Fig 3: Prediction Confidence Map
Probability heatmap ranging [0 - 1]



Fig 4: Blended Spatial Overlay
SegFormer prediction mask overlaid on original RGB

3. Geotechnical Analysis Metrics

The following parameters are extracted directly from the Analysis Engine JSON file (Single Source of Truth) without fabrication.

3.1 Coherence Analysis

Metric	Value	Metric	Value
Mean Coherence	0.5324	Variance	0.0244
Median Coherence	0.5321	Std Deviation	0.1562
Minimum Coherence	0.126	Range	0.7938
Maximum Coherence	0.9197	Skewness	-0.0722
Quartile Q25	0.4214	Kurtosis	-0.6528
Quartile Q50	0.5321	Low Coherence % (<0.3)	7.74%
Quartile Q75	0.6521	Medium Coherence % (0.3-0.7)	75.87%
		High Coherence % (>0.7)	16.39%

3.2 Phase Analysis

Metric	Value	Metric	Value
Mean Phase	0.6368	Std Deviation	0.9903
Median Phase	0.6541	Variance	0.9806
Minimum Phase	-3.1292	Entropy	4.5227
Maximum Phase	3.1247	Energy	25.4925
Range	6.2539	Gradient Mean	0.2006
Skewness	-1.6565	Gradient Std	0.4679
Kurtosis	4.4829		

3.3 Segmentation Analysis

Parameter	Monitored Value	Statistical Explanation
Ground Truth Area	790 px	Annotated reference landslide area
Predicted Area	896 px	Total landslide area detected by SegFormer
Difference	106 px	Spatial mismatch area
Area Percentage	8.96%	Ratio of predicted slide to total monitored block
Average Probability	0.0904	Mean segmentation confidence score
Maximum Probability	1	Peak confidence classification pixel
Minimum Probability	0	Lowest active boundary probability
Dice / F1 Score	0.8873	Similarity coefficient (Ground Truth vs Prediction)
IoU (Jaccard)	0.7974	Intersection over Union spatial overlap metric
Precision	0.8348	Accuracy of predicted positive boundaries
Recall / Sensitivity	0.9468	Ability to locate true landslide boundaries
Specificity	0.9839	Accuracy of non-landslide background pixels
Overall Accuracy	0.981	Percentage of correctly classified pixels

3.4 Shape Analysis

Metric	Value	Metric	Value
Connected Components	2	Convex Area	609 px
Largest Component	618 px	Solidity Ratio	0.9425
Smallest Component	278 px	Aspect Ratio	0.6857
Average Component Area	448 px	Circularity	0.7573
Perimeter Length	97.598	Shape Complexity	16.5947
Bounding Box [x,y,w,h]	[4, 51, 24, 35]	Centroid Coords	(14.7, 69.7)

3.5 Confidence Analysis

Parameter	Observed Metric	Geotechnical Interpretation
Average Probability	0.0904	System's global classification confidence
Maximum Probability	1	Peak signal classification probability
Minimum Probability	0	Active hazard boundary fringe probability
Confidence Variance	0.0762	Spatial deviation of prediction probability
Confidence Entropy	-1739.0239	Information entropy (prediction uncertainty)

3.6 Severity Assessment

Severity Assessment Category
Computed Severity Index: 0.2976
Hazard Risk Level: LOW
Assessment Confidence: LOW

4. Integrated Geotechnical Assessment

4.1 Technical Analysis & Engineering Interpretation

Geodetic InSAR & Coherence Analysis

- **Mean Coherence:** 0.5324468612670898
- **Median Coherence:** 0.5320956707000732
- **Coherence Range:** 0.7937504053115845 [Min: 0.1259699910879135, Max: 0.9197204113006592]
- **Standard Deviation:** 0.15622876584529877
- **Low Coherence Zone (<0.30):** 7.74%
- **Medium Coherence Zone (0.30-0.70):** 75.87%
- **High Coherence Zone (>0.70):** 16.39%

Interferometric Phase & Gradient Analysis

- **Phase Mean / Median:** 0.6368029117584229 / 0.6541308164596558
- **Phase Entropy:** 4.522673892975114
- **Phase Energy:** 25.49247392852968
- **Gradient Magnitude Mean:** 0.20062404870986938 (Standard Deviation: 0.46788567304611206)

Segmentation & Morphological Shape Analysis

- **Predicted Core Area:** 896 pixels (8.96% spatial coverage)
- **Morphological Complexity (Perimeter² / Area):** 16.5947134462359 (Perimeter: 97.5979790687561 px)
- **Circularity / Solidity:** 0.7572514376243973 / 0.9425287356167072
- **Connected Components:** 2 (Largest: 618 px, Smallest: 278 px)

4.2 Plain Language Summary

Ground stability measurements from satellite radar indicate a sliding motion occurring on the slope located in the **Vanoise** region. Our automated AI model has identified active shifting boundaries covering 8.96% of the monitored area. While these slow slips are common in mountainous regions, the steep structure combined with high soil moisture poses a ****LOW**** landslide risk. Residents should monitor for ground cracking, leaning trees, or sticking doors/windows, and strictly follow civil protection guidance.

5. Mitigation & Monitoring Recommendations

5.1 Civil Protection Recommendations

- 1. Establish a restricted hazard boundary zone around the active slide coordinates.
- 2. Direct local emergency services to prepare emergency plans based on the **LOW** hazard status.
- 3. Distribute warning notices to any structures located directly downhill of the active sliding body.

5.2 Engineering Recommendations

- 1. Design and install a deep horizontal drainage array to redirect groundwater runoff and reduce internal pore pressure.
- 2. Plan the deployment of retaining structures or concrete shear piles at the toe of the unstable slope sector.
- 3. Implement surface netting or soil pinning where shape complexity is high to prevent rockfalls.

5.3 Monitoring Recommendations

- 1. Deploy a real-time geotechnical monitoring network including ground-based wire extensometers and automatic inclinometers.
- 2. Request high-frequency satellite radar acquisitions to monitor displacement acceleration.
- 3. Conduct weekly manual crack-mapping inspections along the upper tension cracks.

6. Appendix & System Metadata

- **Analysis Execution Date:** 2026-07-06 12:43:24
- **VLM Pipeline:** Mock Geotechnical Vision-Language Model v1.0
- **Primary Segmenter:** SegFormer-B2
- **Image Inputs Loaded:** RGB=True, Prediction=False, Heatmap=False, Overlay=False

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